

Time Series Analysis

Seminar 8: Long memory and ARFIMA



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Acronyms used in this material

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- **ARIMA**: AutoRegressive Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **CPI**: Consumer Price Index
- **DFA**: Detrended Fluctuation Analysis
- **EODHD**: EOD Historical Data (market-data provider)
- **FIGARCH**: Fractionally Integrated GARCH
- **FRED**: Federal Reserve Economic Data
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GPH**: Geweke–Porter-Hudak (estimator)
- **HICP**: Harmonised Index of Consumer Prices
- **LW**: Local Whittle (estimator)
- **MA**: Moving Average
- **ML**: Machine Learning
- **OLS**: Ordinary Least Squares
- **RV**: Realised Volatility
- **SE**: Standard Error
- **TSA**: Time Series Analysis
- **US**: United States

Today's Question and Seminar Structure

- The question of the seminar
  - **does a shock to a series fade at an exponential rate (ARMA) or much more slowly (long memory), and how do we measure the difference?**
- Results to obtain (practice only: nothing is handed in or graded)
  - the answers to the A exercises
  - a figure and a table for each B exercise
  - the corrected AI answer of C2
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- The exercises, their types and their order: the exercise map (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	weights of $(1-L)^d$, ACF of ARFIMA$(0,d,0)$, ranges of d	Solved, Proposed	A1
A3, A4	R/S and H by hand; a GPH slope by hand	Solved, Proposed	A3
A5, A6	an ARFIMA forecast; GARCH and FIGARCH weights	Solved, Proposed	A5
B1, B2	the Nile and its break; Romanian inflation	Solved, Proposed	B1
B3, B4	S&P 500 volatility; BET and EUR/RON	Solved, Proposed	B3
C1, C2	US inflation and regimes; what is wrong in an AI answer?	Proposed	B1, B2

- Types: **A** calculations on paper; **B** real data, in the notebook; **C** open questions, an AI critique
- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in the seminar

Data used

Series	Source	Frequency	Period
Nile flow at Aswan	statsmodels (nile)	yearly	1871–1970
HICP, Romania	Eurostat (prc_hicp_minr)	monthly, 2015 = 100	2005–2026
CPI, United States	FRED (CPIAUCSL)	monthly, SA	1947–2026
S&P 500, BET	EODHD	daily, close	2000–2026
EUR/RON	BNR reference rate	daily	2005–2026

- Romanian inflation: monthly rate $100\Delta \ln P_t$ minus its monthly means (seasonal adjustment), and the 12-month rate $100(\ln P_t - \ln P_{t-12})$
- In the notebook: `gph`, `local_whittle`, `hurst_rs`, `arfima_exact_ml`, `monthly_rv`; no account or key is needed

Prerequisites for Today (1/22): autocorrelation and memory

- **Autocovariance** and **autocorrelation** of a stationary series x_t at lag k

$$\gamma(k) = \mathrm{Cov}(x_t,\ x_{t-k}), \qquad \rho(k) = \frac{\gamma(k)}{\gamma(0)}$$

  - $\gamma(0) = \mathrm{Var}(x_t)$; $\rho(k) \in [-1, 1]$: the correlation between two values k periods apart
- **Sample autocorrelation** (the ACF of the data, Chapter 1)

$$\hat{\rho}(k) = \frac{\sum_{t=k+1}^{T}(x_t - \bar{x})(x_{t-k} - \bar{x})}{\sum_{t=1}^{T}(x_t - \bar{x})^2}$$

  - T: the number of observations; $\bar{x}$: the sample mean; band of no correlation: $\pm 1.96/\sqrt{T}$
- **Memory**: how fast $\rho(k)$ goes to zero as the lag k grows
  - fast decay: a shock is forgotten quickly; slow decay: the distant past still matters

Prerequisites for Today (2/22): short memory and long memory

- **Short memory** (ARMA models, Chapter 2): exponential decay

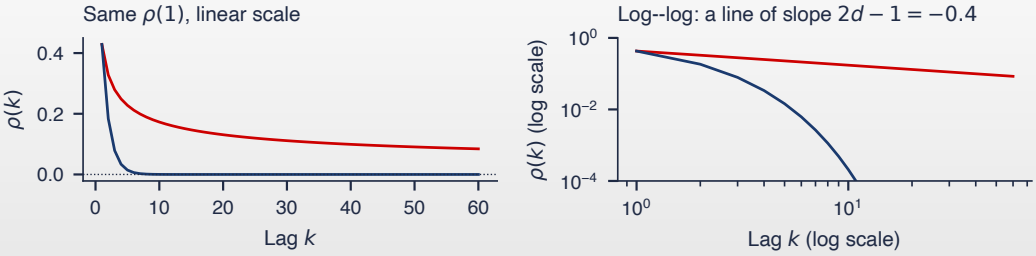
$$|\rho(k)| \leq C\, r^k, \qquad 0 < r < 1$$

  - $C > 0$: a constant; r: the decay factor per lag; AR(1): $\rho(k) = \phi^k$, so $r = |\phi|$
  - the autocorrelations are summable: $\sum_k |\rho(k)| < \infty$
- **Long memory**: hyperbolic (power-law) decay

$$\rho(k) \sim C\, k^{2d-1}, \qquad 0 < d < \tfrac{1}{2}$$

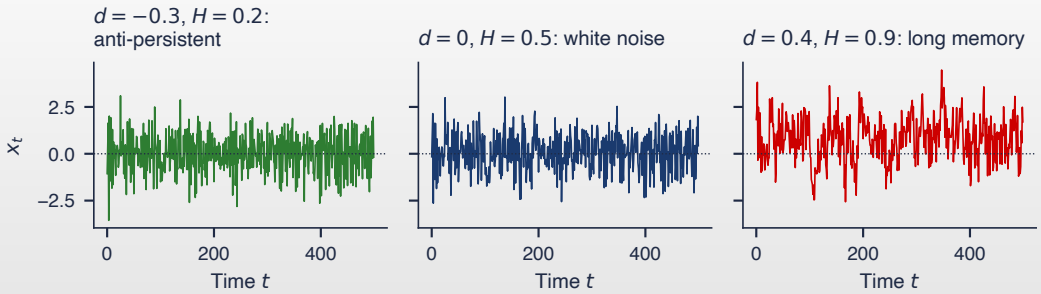
  - d: the **memory parameter**; $\sim$: the ratio of the two sides tends to 1 as $k \to \infty$
  - the exponent $2d - 1$ lies in $(-1, 0)$: the decay is so slow that $\sum_k \rho(k) = \infty$
- **Hurst exponent** $H = d + 1/2$
  - $H = 0.5$: no memory; $H > 0.5$: persistence; $H < 0.5$: anti-persistence
  - Hurst found $H \approx 0.7$ for the yearly flow of the Nile Hurst (1951)

Prerequisites for Today (3/22): the two decays in a chart



- Left: ARFIMA$(0, 0.3, 0)$ and an AR(1) with the same $\rho(1) = 0.429$
  - at lag 10: $\rho(10) = 0.173$ for ARFIMA, but only $0.429^{\mathbf{10}} = 0.0002$ for the AR(1)
- Right: the same curves on log–log axes
  - $\log \rho(k) \approx \log C + (2d - 1) \log k$: a power law is a straight line of slope $2d - 1 = -0.4$
  - the exponential decay bends down and leaves the chart after about 10 lags

Prerequisites for Today (4/22): simulated paths for three values of d



- The same shocks ε_t (white noise, variance 1), filtered with three values of d (the model is defined on the next slides)
- $d = -0.3$ ($H = 0.2$): **anti-persistent**
  - a value above the mean tends to be followed by one below: $\rho(1) < 0$, the path looks rougher than white noise
- $d = 0.4$ ($H = 0.9$): **persistent**
  - long excursions above or below the mean, which look like trends or changes of level, although the mean is constant

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Prerequisites for Today (5/22): the lag operator and fractional differencing

- **Lag operator** L: $Lx_t = x_{t-1}$, $L^k x_t = x_{t-k}$
  - first difference: $(1-L)x_t = x_t - x_{t-1}$, the case $d = 1$
- **Fractional difference** $(1-L)^d$ for a real d: the binomial series

$$(1-L)^d x_t = \sum_{k=0}^{\infty} \pi_k \, x_{t-k}, \qquad \pi_0 = 1, \quad \pi_k = \pi_{k-1} \, \frac{k-1-d}{k}$$

  - π_k: the weight of the value x_{t-k}; $d = 1$ gives $\pi_1 = -1$ and $\pi_k = 0$ for $k \geq 2$ (the usual difference)
  - for $0 < d < 1$: every $\pi_k < 0$ ($k \geq 1$) and $|\pi_k|$ decreases slowly, like k^{-1-d}: all past values count
- Example, $d = 0.4$: $\pi_1 = -0.4$, $\pi_2 = -0.4 \cdot 0.6/2 = -0.12$, $\pi_3 = -0.12 \cdot 1.6/3 = -0.064$

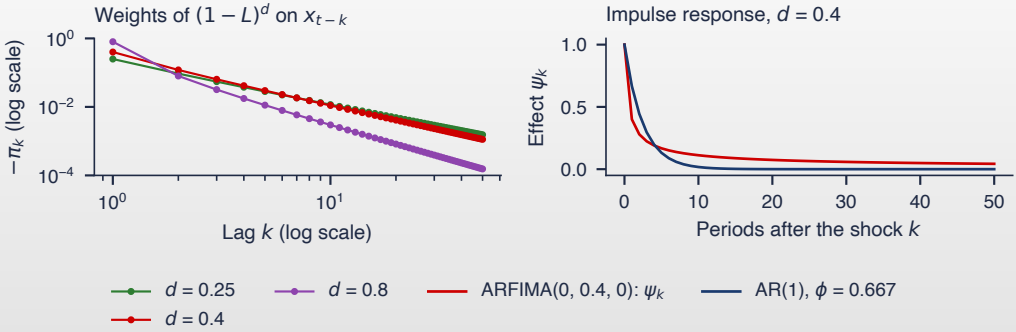
Prerequisites for Today (6/22): the impulse response ψ_k

- **Inverse filter**: the series as a sum of past shocks (the MA(∞) form)

$$x_t = (1-L)^{-d}\varepsilon_t = \sum_{k=0}^{\infty} \psi_k \, \varepsilon_{t-k}, \qquad \psi_0 = 1, \quad \psi_k = \psi_{k-1} \, \frac{k-1+d}{k}$$

  - ψ_k: the effect on x_t of a unit shock that happened k periods ago
- Example, $d = 0.4$: $\psi_1 = 0.4$, $\psi_2 = 0.4 \cdot 1.4/2 = 0.28$, $\psi_3 = 0.28 \cdot 2.4/3 = 0.224$
  - for $0 < d < 1/2$, ψ_k decreases like k^{d-1}: slowly, but towards 0, so the series is stationary
- For an AR(1), $\psi_k = \phi^k$: the effect of a shock dies out exponentially

Prerequisites for Today (7/22): fractional weights and impulse responses in a chart



- Left: the weights $-\pi_k$ of $(1-L)^d$ on log–log axes are almost straight lines: a power law in k
- Right: for $d = 0.4$, $\psi_{10} = 0.112$, against $0.667^{10} = 0.017$ for an AR(1) with the same $\rho(1) = 0.667$

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Prerequisites for Today (8/22): the ARFIMA model

- **ARFIMA**(p, d, q): an ARMA model for the fractionally differenced series Granger and Joyeux (1980), Hosking (1981)

$$\phi(L)\,(1-L)^d\,(x_t-\mu)=\theta(L)\,\varepsilon_t$$

  - $\phi(L)=1-\phi_1 L-\cdots-\phi_p L^p$, $\theta(L)=1+\theta_1 L+\cdots+\theta_q L^q$: the AR and MA polynomials (Chapter 2)
  - μ: the mean; ε_t: white noise with variance σ^2; $d=0$: ARMA; $d=1$: ARIMA$(p,1,q)$

Range of d	Properties
$d \le -1/2$	not invertible
$-1/2 < d < 0$	stationary, invertible, anti-persistent
$d = 0$	short memory (ARMA)
$0 < d < 1/2$	stationary, invertible, long memory
$1/2 \le d < 1$	not stationary, but mean-reverting: a shock still dies out
$d = 1$	unit root: a shock is permanent

- For $1/2 \le d < 3/2$, difference once: $(1-L)x_t$ is ARFIMA with memory parameter $d-1$

Prerequisites for Today (9/22): ARFIMA$(0, d, 0)$ in numbers

- **Autocorrelations** by recursion

$$\rho(1) = \frac{d}{1-d}, \qquad \rho(k) = \rho(k-1)\,\frac{k-1+d}{k-d}, \quad k \geq 2$$

  - example $d = 0.3$: $\rho(1) = 0.3/0.7 = 0.429$, $\rho(2) = 0.429 \cdot 1.3/1.7 = 0.328$
  - for large k: $\rho(k) \approx \frac{\Gamma(1-d)}{\Gamma(d)}\, k^{2d-1}$, the constant C of the long-memory definition

- **Variance**

$$\gamma(0) = \sigma^2\,\frac{\Gamma(1-2d)}{\Gamma(1-d)^2}$$

  - $\Gamma(z) = \int_0^\infty u^{z-1}e^{-u}\,du$: the gamma function; $\Gamma(n) = (n-1)!$ for integers $n \geq 1$; $\Gamma(1/2) = \sqrt{\pi}$
  - $d > 0$: $\gamma(0) > \sigma^2$, the memory accumulates past shocks; $\gamma(0) \to \infty$ as $d \to 1/2$

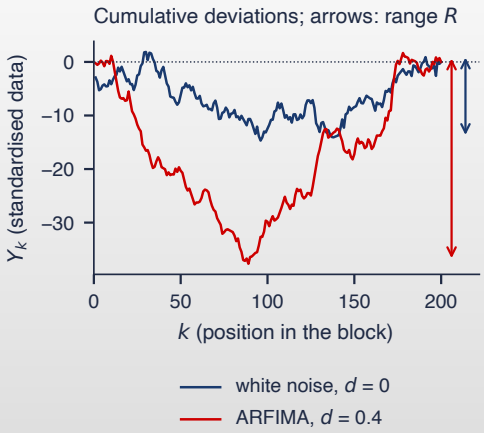
Prerequisites for Today (10/22): the one-step forecast

- From $(1-L)^d(x_{T+1}-\mu)=\varepsilon_{T+1}$, replace the unknown shock ε_{T+1} by its mean, 0

$$\hat{x}_{T+1}=\mu-\sum_{k\geq 1}\pi_k\left(x_{T+1-k}-\mu\right)$$

  - x_T: the last observation; x_{T+1-k}: the value $k-1$ periods before it; in practice the sum stops at $k=T$
- Reading the formula for $0<d<1$
  - the weights $-\pi_k$ are positive: each past deviation from μ pushes the forecast in its own direction
  - the weights decrease slowly, so old deviations still count
- Comparison: AR(1), $\hat{x}_{T+1}=\mu+\phi(x_T-\mu)$
  - only the last value enters the forecast; ϕ: the AR(1) coefficient

Prerequisites for Today (11/22): the R/S statistic



- For a block $x_1, \ldots, x_n$ with mean $\bar{x}$ Hurst (1951):

$$Y_k = \sum_{i=1}^{k} (x_i - \bar{x}), \quad k = 1, \ldots, n$$

  - Y_k: the cumulative deviation from the mean; $Y_n = 0$

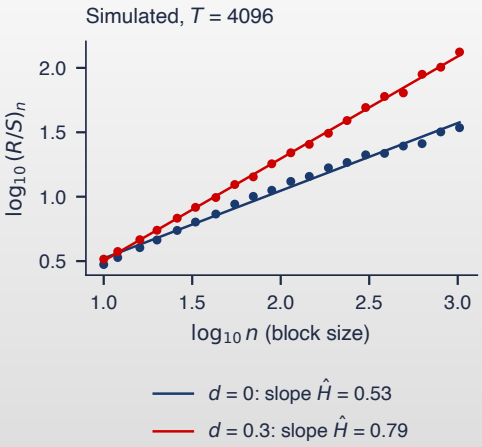
- **Range** and **standard deviation**

$$R = \max_k Y_k - \min_k Y_k, \quad S = \sqrt{\tfrac{1}{n} \textstyle\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

  - R/S: the range of the cumulative deviations, in units of S; it grows like n^H

- Chart: a persistent block has long runs of the same sign
  - Y_k drifts far from 0, so R is large (39 against 17 for white noise)

Prerequisites for Today (12/22): H from a log–log regression



- Split the series in blocks of size n and average R/S over the blocks: $(R/S)_n$

$$\log_{10}(R/S)_n \approx a + H \log_{10} n$$

  - repeat for several n; $\hat{H}$ = the OLS slope of $\log_{10}(R/S)_n$ on $\log_{10} n$; $\hat{d} = \hat{H} - 0.5$
  - a: the intercept; OLS: ordinary least squares
- Chart: white noise gives $\hat{H} = 0.53$, slightly above 0.5
  - R/S is biased upwards in small blocks; $d = 0.3$ gives $\hat{H} = 0.79 \approx 0.8$
- **DFA** (detrended fluctuation analysis) Peng et al. (1994)
  - cumulate the series, fit a line in each block of size n; $F(n)$: the root mean square of the residuals
  - $F(n) \propto n^H$: $\hat{H}$ = the slope of $\log F(n)$ on $\log n$

Prerequisites for Today (13/22): the periodogram and the spectral density

- **Fourier frequencies** $\lambda_j = 2\pi j/T$, $j = 1, \ldots, \lfloor T/2 \rfloor$
  - λ_j in radians per period; it corresponds to a cycle of T/j periods; $\lfloor \cdot \rfloor$: the integer part
  - low frequencies (small j): long cycles; this is where long memory shows
- **Periodogram**: how much of the variance is in cycles of frequency λ_j

$$I(\lambda_j) = \frac{1}{2\pi T} \left| \sum_{t=1}^{T} (x_t - \bar{x})\, e^{-i\lambda_j t} \right|^2$$

  - i: the imaginary unit; $I(\lambda_j)$ is a noisy estimate of the spectral density $f(\lambda_j)$
- **Long memory** = a pole of the spectral density at frequency zero

$$f(\lambda) \sim G\, \lambda^{-2d} \quad \text{as } \lambda \to 0$$

  - $G > 0$: a constant; ARFIMA$(0, d, 0)$: $f(\lambda) = \frac{\sigma^2}{2\pi} \left(4 \sin^2(\lambda/2)\right)^{-d}$, and $4 \sin^2(\lambda/2) \approx \lambda^2$ for small λ

Prerequisites for Today (14/22): the GPH estimator

- **GPH** Geweke and Porter-Hudak (1983): take logs of $I(\lambda_j) \approx f(\lambda_j)$ at the m lowest frequencies: a linear regression in d

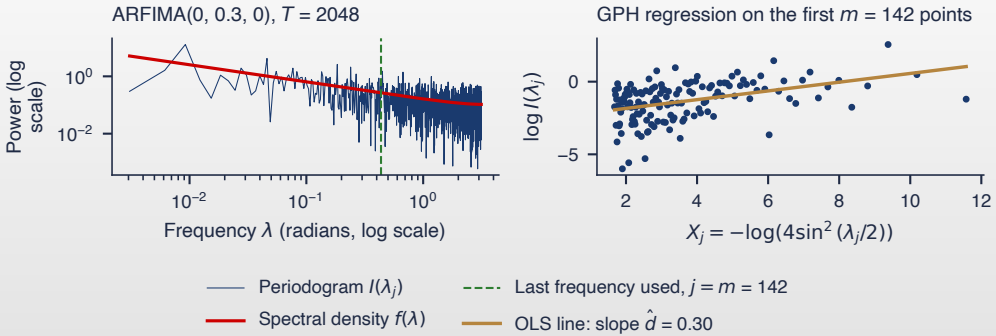
$$\log I(\lambda_j) = c + d\, X_j + u_j, \qquad X_j = -\log\left(4\sin^2(\lambda_j/2)\right), \quad j = 1, \ldots, m$$

  - c: the intercept; u_j: the error; m: the **bandwidth**, the number of low frequencies used
- The estimate is the OLS slope, with $y_j = \log I(\lambda_j)$

$$\hat{d} = \frac{\sum_{j=1}^{m}(X_j - \bar{X})(y_j - \bar{y})}{\sum_{j=1}^{m}(X_j - \bar{X})^2}, \qquad \mathrm{SE}(\hat{d}) = \frac{\pi}{\sqrt{24m}}$$

  - $\bar{X}$, $\bar{y}$: the means over the m frequencies; SE: the asymptotic standard error
- **Test** of H_0: $d = 0$ (no long memory)
  - $t = \hat{d}/\mathrm{SE}(\hat{d})$, compared with the Normal distribution: reject at 5% if $|t| > 1.96$
  - usual bandwidth $m = \lfloor T^{0.65} \rfloor$; a small m: less bias from short-run dynamics, but more noise

Prerequisites for Today (15/22): the periodogram and GPH in a chart



- Left: the periodogram of a simulated ARFIMA$(0, 0.3, 0)$ ($T = 2048$) scatters around $f(\lambda)$, which rises towards $\lambda = 0$
- Right: the GPH regression on the first $m = \lfloor 2048^{\mathbf{0.65}} \rfloor = 142$ frequencies gives $\hat{d} = 0.30$ (SE $= \pi/\sqrt{24 \cdot 142} = 0.054$)

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Prerequisites for Today (16/22): local Whittle, exact ML and the information criteria

- **Local Whittle** (LW) Robinson (1995): $\hat{d}$ minimises, over the same m frequencies,

$$R(d) = \log\Big(\frac{1}{m}\sum_{j=1}^{m}\lambda_j^{2d}\, I(\lambda_j)\Big) - \frac{2d}{m}\sum_{j=1}^{m}\log\lambda_j$$

  - a likelihood that uses only $f(\lambda) \approx G\lambda^{-2d}$ near zero; $\mathrm{SE}(\hat{d}) = 1/(2\sqrt{m})$, smaller than for GPH
- **Exact maximum likelihood** (ML) Sowell (1992)
  - maximises the full Normal likelihood L of ARFIMA(p, d, q) over d, the ϕ_i, the θ_j and σ^2
  - uses all frequencies: more precise if the model is right, biased if the short-run part is wrong
- **Information criteria**: the model with the smallest value wins

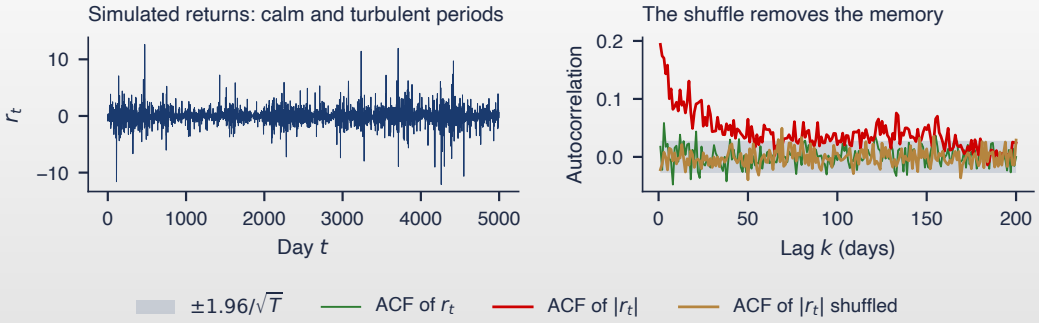
$$\mathrm{AIC} = -2\ln L + 2k, \qquad \mathrm{BIC} = -2\ln L + k\ln T$$

  - $\ln L$: the maximised log-likelihood; k: the number of estimated parameters; BIC penalises extra parameters more

Prerequisites for Today (17/22): memory in volatility and the shuffle test

- Returns $r_t = 100 \ln(P_t / P_{t-1})$ are almost uncorrelated, but their size $|r_t|$ has long memory Ding, Granger and Engle (1993)
  - P_t: the closing price of day t; monthly **realised volatility** $\sqrt{RV_m}$, with $RV_m = \sum_{t \in m} r_t^2$ (the days t of month m); we model $\log RV_m$
- **Shuffle test**: permute the days at random and re-estimate d
  - the permutation keeps the distribution of the values and destroys their order: real memory gives $\hat{d} \approx 0$ after shuffling
- Why: memory in $|r_t|$ means that calm and turbulent periods last (volatility clustering)
  - it concerns the size of the moves, not their sign: the direction stays unpredictable

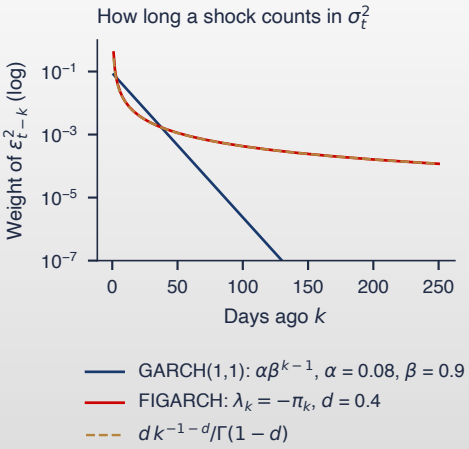
Prerequisites for Today (18/22): the shuffle test in a chart



⊡ Simulated: ACF of $|r_t|$ positive up to lag 200, LW $\hat{d} = 0.28$; after shuffling, inside the band, $\hat{d} = -0.04$

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Prerequisites for Today (19/22): GARCH and FIGARCH weights

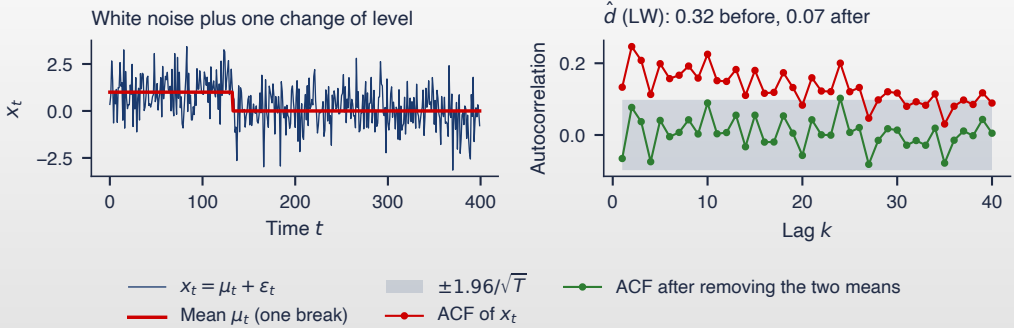


- **GARCH(1,1)** (Chapter 5): $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$
  - σ_t^2: the conditional variance of day t; ε_t: the shock; $\omega > 0$, $\alpha, \beta \geq 0$: the parameters
  - substituting σ_{t-1}^2, σ_{t-2}^2, ... again and again (ARCH(∞) form):
$$\sigma_t^2 = \frac{\omega}{1-\beta} + \alpha \sum_{k \geq 1} \beta^{k-1} \varepsilon_{t-k}^2$$
  - weight of ε_{t-k}^2: $\alpha\beta^{k-1}$, exponential
- **FIGARCH**$(0, d, 0)$ Baillie, Bollerslev and Mikkelsen (1996): $\sigma_t^2 = \omega + [1 - (1-L)^d]\,\varepsilon_t^2$
  - weights $\lambda_k = -\pi_k \approx d\,k^{-1-d}/\Gamma(1-d)$: hyperbolic
  - chart, $k = 100$: $2.4 \cdot 10^{-6}$ (GARCH) against $4.3 \cdot 10^{-4}$ (FIGARCH)

Prerequisites for Today (20/22): spurious long memory

- A short-memory series with a change of level: $x_t = \mu_t + \varepsilon_t$, where the mean μ_t changes once (a **break**)
  - the ACF stays positive at long lags and $\hat{d} > 0$, although there is no memory Diebold and Inoue (2001)
  - remedy: estimate the mean of each regime, subtract it and re-estimate d
- **Monte Carlo band**: simulate many i.i.d. series of the same length, estimate d on each
  - the 2.5% and 97.5% quantiles of these $\hat{d}$: the 95% band of $\hat{d}$ when there is no memory; i.i.d.: independent and identically distributed
- An estimate $\hat{d}$ outside the band: evidence of memory, or of a break; inside: compatible with no memory

Prerequisites for Today (21/22): a break that looks like long memory



- Left: white noise whose mean falls by 1 after one third of the sample ($T = 400$)
- Right: the ACF stays above the band for 40 lags and LW gives $\hat{d} = 0.32$; after removing the two means, $\hat{d} = 0.07$

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Prerequisites for Today (22/22): monthly and 12-month inflation rates

- P_t: the price index of month t; **monthly rate** $\pi_t^m = 100\,\Delta \ln P_t = 100(\ln P_t - \ln P_{t-1})$
  - seasonal adjustment by monthly means: subtract from each month (January, February, ...) its average over all years
- The **12-month rate** is the sum of the last 12 monthly rates

$$100(\ln P_t - \ln P_{t-12}) = \sum_{j=0}^{11} \pi_{t-j}^m$$

  - two consecutive 12-month rates share 11 monthly rates (overlapping sums)
  - even for i.i.d. monthly rates, the 12-month rate has $\rho(k) = (12-k)/12$ for $k < 12$: a strong autocorrelation built by the definition

A1: fractional weights and the ACF [Solved]

Task

- A series is ARFIMA$(0, 0.25, 0)$ with $\sigma^2 = 1$.
- 1. Compute $\pi_1, \ldots, \pi_4$ of $(1 - L)^{0.25}$.
- 2. Compute $\rho(1)$ and $\rho(2)$, and give H and the class of the process.
- 3. Compare $\rho(5)$ with that of an AR(1) with the same $\rho(1)$.
- Report: four weights, two autocorrelations, H, two values of $\rho(5)$.

Solution

- 1. $\pi_1 = -0.25$; $\pi_2 = -0.25 \cdot 0.75/2 = -0.0938$; $\pi_3 = -0.0938 \cdot 1.75/3 = -0.0547$; $\pi_4 = -0.0547 \cdot 2.75/4 = -0.0376$
- 2. $\rho(1) = 0.25/0.75 = 0.333$; $\rho(2) = 0.333 \cdot 1.25/1.75 = 0.238$; $H = 0.75$: stationary, long memory
- 3. ARFIMA: $\rho(5) = 0.151$; AR(1): $0.333^5 = 0.0041$, about 37 times smaller

Back Exercise list

A2: which range is d in? [Proposed]

Solution discussed in class

Task

- Two series: (a) ARFIMA$(0, 0.6, 0)$; (b) ARFIMA$(0, -0.3, 0)$. Model: A1.
- 1. Compute π_1 and π_2 of $(1 - L)^d$ for each.
- 2. Say whether each series is stationary, invertible and mean-reverting.
- 3. For (b) compute $\rho(1)$ and $\rho(2)$; for (a) compute $\rho(1)$ of the first difference.
- Report: four weights, two classifications, three autocorrelations.

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A3: R/S by hand and H from three points [Solved]

Task

- Block: $x = (0.6, 0.4, 0.5, 0.3, -0.4, -0.6, -0.2, -0.5)$. An R/S plot gives $(R/S)_n = 4.1;\ 9.6;\ 22.4$ for $n = 16, 64, 256$.
- 1. Compute $\bar{x}$, the cumulative deviations Y_k, R, S and R/S.
- 2. Compute H as the slope of $\log_{10}(R/S)_n$ on $\log_{10} n$, and $d = H - 0.5$.
- Report: R/S, H, d and one sentence.

Solution

- 1. $\bar{x} = 0.0125$; Y_k: 0.5875, 0.9750, 1.4625, 1.7500, 1.3375, 0.7250, 0.5125, 0; $R = 1.750$, $S = 0.457$, $R/S = 3.83$
- the signs are grouped (four positive, then four negative): the range is large, as for a persistent series
- 2. $\log_{10} n$: 1.204, 1.806, 2.408; $\log_{10}(R/S)$: 0.613, 0.982, 1.350; slope $H = 0.61$, $d = 0.11$: moderate persistence

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A4: a GPH slope by hand [Proposed]

Solution discussed in class

Task

- For $m = 4$ frequencies: $X_j = 0.5;\ 1.6;\ 2.3;\ 3.0$ and $\log I(\lambda_j) = 1.2;\ 1.8;\ 2.0;\ 2.5$. Model: A3.
- 1. Compute $\bar{X}$, $\bar{y}$, $\sum(X_j - \bar{X})(y_j - \bar{y})$ and $\sum(X_j - \bar{X})^2$.
- 2. Compute $\hat{d}$ and its standard error (Prerequisites for Today (14/22)).
- 3. Test H_0: $d = 0$ at 5% and comment on the size of m.
- Report: four sums, $\hat{d}$, its SE, a decision and one sentence.

A5: a one-step ARFIMA forecast [Solved]

Task

- ARFIMA$(0, 0.4, 0)$ with $\mu = 2$; the last four values: $x_T = 3.0$, $x_{T-\mathbf{1}} = 2.5$, $x_{T-\mathbf{2}} = 2.8$, $x_{T-\mathbf{3}} = 1.5$.
- 1. Write the first four weights $-\pi_k$ of $(1 - L)^{\mathbf{0.4}}$.
- 2. Compute $\hat{x}_{T+\mathbf{1}}$ using these four lags.
- 3. Compare with an AR(1) with the same $\rho(1)$.
- Report: four weights, two forecasts and one sentence.

Solution

- 1. $-\pi_k$: 0.4; 0.12; 0.064; 0.0416
- 2. $\hat{x}_{T+\mathbf{1}} =$
  $2 + 0.4(1.0) + 0.12(0.5) + 0.064(0.8) + 0.0416(-0.5) =$
  $2 + 0.4000 + 0.0600 + 0.0512 - 0.0208 = 2.490$
- 3. AR(1): $\phi = \rho(1) = 0.4/0.6 = 0.667$,
  $\hat{x}_{T+\mathbf{1}} = 2 + 0.667 \cdot 1.0 = 2.667$; the AR(1) looks only at x_T, ARFIMA spreads the weight over the past (in practice over all T values)

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A6: how long does a volatility shock count? [Proposed]

Solution discussed in class

Task

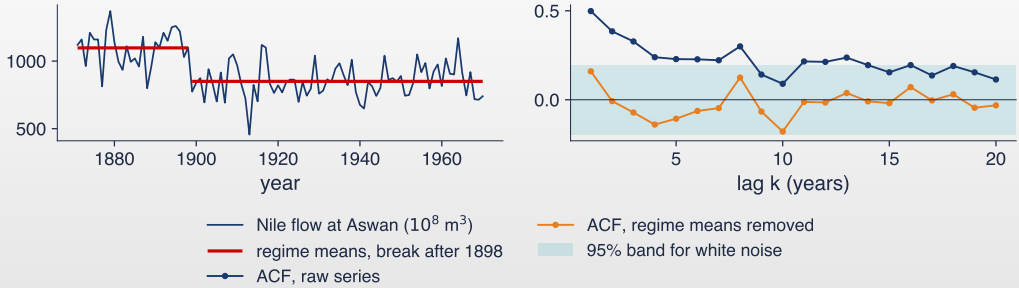
- GARCH(1,1) with $\alpha = 0.10$, $\beta = 0.88$; FIGARCH$(0, d, 0)$ with $d = 0.4$ (weights $\lambda_k = -\pi_k$). Model: A5.
- 1. Compute the weight of ε^2_{t-k} in σ^2_t for $k = 1, 2, 3$ in both models.
- 2. Compute both weights at $k = 20$ and $k = 100$; for FIGARCH use the approximation of λ_k in Prerequisites for Today (19/22), with $\Gamma(0.6) = 1.489$.
- 3. Explain which model keeps a crisis in today's volatility longer.
- Report: ten weights and one sentence.

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B1: does the Nile have long memory? [Solved]

- **Question**: is the persistence of the Nile flow long memory, or the effect of the change in 1898?
- **Inputs**: yearly flow at Aswan, 1871–1970 ($T = 100$)
- **Tasks**
  1. Plot the series and its ACF up to lag 20.
  2. Estimate H by R/S, d by GPH and local Whittle ($m = \lfloor T^{\mathbf{0.65}} \rfloor$) and by exact ML of ARFIMA$(0, d, 0)$.
  3. Subtract the mean of 1871–1898 and the mean of 1899–1970 and repeat step 2.
  4. Interpretation: does the Nile have long memory?
- **Report**: two ACF values, four estimates before and after, one sentence
- Notebook: section B1

B1: solution [Solved]



- Raw: $\hat{\rho}(1) = 0.50$, $\hat{\rho}(10) = 0.09$; $H_{R/S} = 0.78$; GPH 0.44 (SE 0.15), LW 0.40 (SE 0.11), ML 0.36 (SE 0.07); $m = 19$
- Means 1098 and 850; after removing them: $\hat{\rho}(1) = 0.16$; $H_{R/S} = 0.63$; GPH -0.07, LW -0.17, ML 0.05 (SE 0.10)
- Interpretation: for 1871–1970 the evidence of long memory comes from one break; R/S stays above 0.5 because it is biased upwards in short samples

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B2: long memory in Romanian inflation [Proposed]

- **Question**: how persistent is Romanian inflation, and does the way we measure inflation change the answer?
- **Inputs**: monthly HICP, 2005–2026: the monthly rate seasonally adjusted with the monthly means, and the 12-month rate; model: B1
- **Tasks**
  1. Estimate d of the monthly rate by GPH and local Whittle with $m = \lfloor T^a \rfloor$, $a = 0.5;\ 0.65;\ 0.8$.
  2. Fit ARFIMA$(0, d, 0)$ and ARMA(1,1) by exact ML and compare their BIC.
  3. Estimate d of the 12-month rate by local Whittle and compare the two ACFs.
  4. Interpretation: why does the 12-month rate give d close to 1?
- **Report**: six estimates, two BIC values, one d, two sentences
- Notebook: section B2

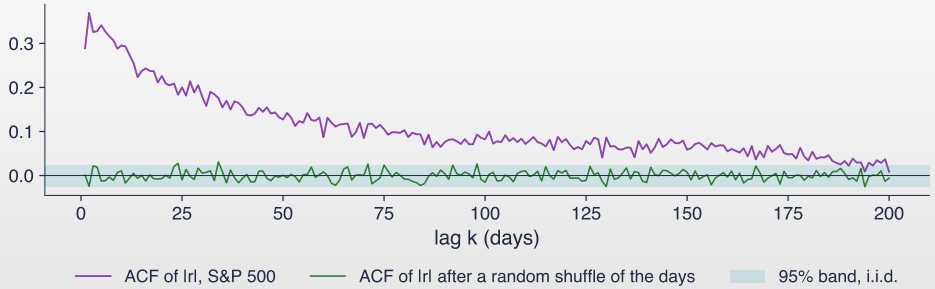
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B3: memory in S&P 500 volatility [Solved]

- **Question**: where is the memory of stock returns: in the returns or in their size?
- **Inputs**: daily log returns of the S&P 500 since 2000 and their monthly realised volatility
- **Tasks**
  1. Estimate d of r_t and of $|r_t|$ by local Whittle and GPH.
  2. Shuffle the days at random and re-estimate d of $|r_t|$; plot both ACFs of $|r_t|$.
  3. Estimate d of the monthly log realised volatility.
  4. Interpretation: what does the shuffle test show?
- **Report**: eight estimates, the chart and one sentence
- Notebook: section B3

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B3: solution [Solved]



- $T = 6\,714$ days; r_t: LW -0.01, GPH 0.02 (SE 0.03, 0.04); $|r_t|$: LW 0.53, GPH 0.56
- Shuffled $|r_t|$: LW -0.03, GPH -0.03; ACF of $|r_t|$: 0.289 (lag 1), 0.127 (lag 50), 0.009 (lag 200); log RV (321 months): LW 0.41, GPH 0.37
- Interpretation: the shuffle keeps every daily value and destroys their order; the memory disappears, so it lives in the clustering of calm and turbulent periods, not in the distribution of returns

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B4: BET and EUR/RON [Proposed]

- **Question**: do the BET and the EUR/RON rate show the same memory, and is it stable before and after the 2008 crisis?
- **Inputs**: daily log returns of the BET (since 2000) and of the EUR/RON reference rate (since July 2005); model: B3
- **Tasks**
  1. Estimate d of r_t and $|r_t|$ by local Whittle on the full sample.
  2. Repeat on the subsamples up to 2007 and 2010–2026.
  3. Compare the estimates for r_t with the 95% Monte Carlo band of i.i.d. series of 1900 days.
  4. Interpretation: is the full-sample memory of BET returns real?
- **Report**: twelve estimates, one band and two sentences
- Notebook: section B4

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C1: US inflation and monetary regimes [Proposed]

- **Question**: is the long memory of US inflation a property of inflation, or of the changes in monetary policy?
- **Inputs**: US monthly CPI inflation, annualised, 1947–2026; regimes 1947–1984, 1985–2019, 2020–2026; models: B1, B2
- **Tasks**
  1. Estimate d by local Whittle on the full sample and in each regime.
  2. Remove the three regime means and re-estimate d.
  3. Propose one more check (a break test, a Markov-switching model, a forecast comparison).
  4. Interpretation: can a constant d describe 80 years of inflation?
- **Report**: five estimates, one check and a project plan
- Notebook: section C1

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C2: audit an AI answer [Proposed]

- A student asked an AI assistant about the long memory of Romanian inflation and of the S&P 500. The answer:
- `(a) Monthly inflation has d = 0.31, so its Hurst exponent is H = d - 0.5 < 0: it is anti-persistent.`
- `(b) The 12-month rate gives d = 1.00, so Romanian inflation has a unit root.`
- `(c) With d = 0.31 the series is stationary and its shocks die out.`
- `(d) Use statsmodels.tsa.arima.model.ARIMA(x, order=(0, 0.31, 0)) to fit the ARFIMA model.`
- `(e) S&P 500 returns have d = -0.01 and |r| has d = 0.53, so returns are predictable.`
- `(f) A significant GPH estimate proves long memory.`
- **Tasks**
  1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
- **Report**: a list of six verdicts with one line of justification each

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Key takeaways

- $(1-L)^d$: weights by the recursion $\pi_k = \pi_{k-1}(k-1-d)/k$; ARFIMA$(0,d,0)$: $\rho(1) = d/(1-d)$
- $H = d + 1/2$; R/S and GPH are slopes of log–log regressions
- Romanian monthly inflation: $d \approx 0.3$; S&P 500: memory in $|r_t|$, not in r_t
- Breaks can create long memory: the Nile, BET returns, US inflation
- An AI answer is a draft: check H against d, the library, and the data transformation

After the seminar

- Lecture 8 develops each topic of today
  - hyperbolic decay, fractional differencing, ARFIMA, the estimators, forecasting, volatility, spurious memory
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: inflation memory and monetary regimes in Romania
- Reading: Baillie (1996); Graves et al. (2017); Ding, Granger and Engle (1993)

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